

SUPERVISED MACHINE LEARNING
Random Forest Classification

Jean René RANDRIANJAKAMANANA

2024-11-27

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3) Random Forest Classifier of Ensemble

This time we shall use the algorithm of **Random Forest Classification** to classify the datasets of iris.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 import pandas as pd
4
5 from sklearn.ensemble import RandomForestClassifier
6 from sklearn.datasets import load_iris
7 from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
8
9 iris = load_iris()
10 y = iris.target
11 X = iris.data
12
13 modele = RandomForestClassifier()
14 modele.fit(X, y)
15
16 y_pred = modele.predict(X)
17 R2_sc = r2_score(y, y_pred)
18 R2 = modele.score(X, y)
19
20 mse = mean_squared_error(y, y_pred)
21 rmse = mean_squared_error(y, y_pred, squared = False)
22 mae = mean_absolute_error(y, y_pred)
23
24 print(f"\nScore (Error in the prediction): {R2}")
25 print(f"Score (Error in the datasets): {R2_sc}\n")
26
27 print(f"MSE = {mse}")
28 print(f"RMSE = {rmse}")
29 print(f"MAE = {mae}")
30
31 plt.scatter(X[:,0], y, c = 'b', label = 'Real datasets')
32 plt.scatter(X[:,0], y_pred, c = 'r', label = 'Predicition datasets')
33 plt.legend()
34 plt.show()
35 plt.close()
```

Score (Error in the prediction): 1.0
Score (Error in the datasets): 1.0

MSE = 0.0
RMSE = 0.0
MAE = 0.0

